



Analytics Engineer Case Study

Getsafe builds the best insurance! We offer a fully digital insurance experience, easy to use, and loved by customers.

The Data Team develops data products to make sure the company has all the relevant information for making the right decisions and delight our customers.

Below you will find two questions that are related to typical projects carried out by the Data Team. Please approach them with an Analytics Engineering perspective, meaning, we want to build reusable solutions as opposed to one-time analysis.

Part I: Data modeling

The accounting team performs monthly closing, and reports total premiums manually with invoices. On the other side, the Finance team requests us to report the same number from our raw transaction record.

The monthly closing from an accounting perspective is the following:

Accounting		
Party	Month	Premium
berlinre	2025-Jun	1483.22
berlinre	2025-Jul	1098.74
berlinre	2025-Aug	714.06
dronant	2025-Jun	2983.41
dronant	2025-Jul	1780.19
dronant	2025-Aug	1107.07
getland	2025-Jun	1730.31
getland	2025-Jul	1412.28
getland	2025-Aug	660.19
liadigital	2025-Jun	2241.79
liadigital	2025-Jul	1353.47
liadigital	2025-Aug	878.49

On the other hand, you are provided with the dataset attached (Data for Analytics Case Study - Raw Transaction Data.csv)

Your tasks:

1. Please write a dbt model (or models) that generates a data mart from the dataset provided with the following columns:
 - a. party
 - b. month
 - c. premium
2. Describe a process to compare the finance results with the accounting closing information and generate a report on such discrepancies.
3. Are there discrepancies between the two sources? If there are discrepancies, please discuss what could be potential solutions.

Part 2: KPIs calculation

In this part you will help us design and implement a data mart that will be connected to a BI tool for further visualizations. The input is the following dataset:

products_customers

	ACQUISITION DATE	STARTED AT	CHURNED AT	USER ID	PREMIUM	PRODUCT GROUP
1	2025-07-01 15:28:00	2025-07-02 00:00:00		1-088f	49.04	dog
2	2025-11-29 09:19:49	2025-11-30 00:00:00		5-169e	42.78	contents
3	2024-10-03 08:58:17	2024-10-04 00:00:00	2025-01-21 00:00:00	7-86d6	69.92	dental
4	2025-08-02 18:24:08	2025-08-03 00:00:00		0-3594	50.83	dog
5	2024-06-19 21:25:10	2024-06-20 00:00:00		1-18c6	22.13	liability
6	2024-01-09 09:52:50	2024-01-10 00:00:00		9-8903	0.19	liability
7	2025-09-18 12:01:22	2025-09-19 00:00:00		d-604b	32.56	legal
8	2024-09-16 11:30:12	2024-09-17 00:00:00	2025-06-16 00:00:00	6-3842	15.13	liability
9	2025-11-30 21:12:10	2025-01-01 00:00:00		9-918f	83.10	car
10	2025-02-14 12:14:58	2025-02-15 00:00:00	2025-03-17 00:00:00	9-ddea	11.34	legal

The columns in the input dataset are the following:

- Acquisition date: Date when the customer signs their contract with us.
- Started at: Date when the contract starts.
- Churned at: When the customer canceled their contract, this is the last day when they were our customers. If the customer is still active, this value is empty.
- User id: Unique identifier per customer
- Premium: Monthly amount to pay for the insurance products acquired by the customer.
- Product group: Category of the insurance product acquired.

The metrics we want to provide are the following:

- **Number of active customers per day and product group in a given date range.** For example, if a customer starts on Jan 1st and churns on March 31st, the customer is considered active between Jan 1st and March 31st, and is not considered active outside of that period.
- **Premium collected from active customers in a given date range.** If the customer starts on Jan 1st and churns on March 31st, we will collect the premium between Jan 1st and March 31st only.
- **Accumulated acquired premium from the beginning of time until a given date.**

Assumptions:

- The data mart produced should allow generating the metrics in the BI tool with minimal aggregations. The only aggregations to be done in the BI tool are the operations *sum* and *count*.
- The input dataset can become very big, so, consider scalability and flexibility in the date ranges.
- There are processes that can change data retroactively. For example, we might allow a customer to cancel his contract effectively in the past, and we will refund the money they paid. That means that a value that was reported in the past can be different now.

Your task:

Please describe the workflow that would allow us to generate the data mart, covering the assumptions mentioned above. Include the code snippets necessary to make it happen.

You can submit your answers in a format of your choice, in such a way that we can follow your reasoning and see your code :)

Please contact Bastian (b.laubner@hellogetsafe.com) at any time if you have questions on the given topics.

Enjoy!